

GR&R; Study Report

Gauge Repeatability & Reproducibility -- AIAG MSA 4th Edition | 10 Parts | 3 Operators | 3 Trials

Equipment / Gage:	Zeiss Contura CMM SN-4471
Study Performed By:	QE Team
Report Date:	2026-05-10
Study Design:	3 Operators, AIAG Crossed GR&R
Regulatory Reference:	21 CFR 820.72 AIAG MSA 4th Edition

Variation Components (%Study Variation)

Component	Study Variation	% of TV	Assessment
EV -- Equipment (Repeatability)	0.00577	6.2%	Low
AV -- Appraiser (Reproducibility)	0.00490	5.3%	Low
GR&R -- Combined	0.00757	8.2%	Acceptable
PV -- Part Variation	0.09232	99.7%	--
TV -- Total Variation	0.09263	100.0%	--

Assessment column: GR&R; row uses AIAG pass/fail criteria (<=10% Acceptable / 10-30% Marginal / >30% Unacceptable). EV and AV rows show diagnostic level (Low / Moderate / High) to identify the dominant variation source.

Number of Distinct Categories (NDC): **17** -- Adequate: gauge resolves >=5 distinct categories of part variation. [PASS]

Intermediate Calculation Values

Parameter	Symbol	Value	Formula
Grand Mean	X-bar-bar	14.99318	Mean of all measurements
Avg Range (overall)	R-bar-bar	0.00189	Mean of per-operator R-bars
Upper Control Limit (R)	UCL_R	0.00487	D4(2.574) x R-bar-bar
Operator Mean Spread	X-diff	0.00186	max(op X-bar) - min(op X-bar)
Part Range	Rp	0.05699	max(part avg) - min(part avg)
K1 constant (EV)	K1	3.05	5.15 / d2*(3 trials)
K2 constant (AV)	K2	2.7	5.15 / d2*(3 operators)
K3 constant (PV)	K3	1.62	5.15 / d2*(10 parts)

Per-Operator Breakdown

Operator	Grand Mean (X-bar)	Avg Range (R-bar)
Alice	14.99321	0.00176
Bob	14.99409	0.00210

Operator	Grand Mean (X-bar)	Avg Range (R-bar)
Carol	14.99223	0.00182

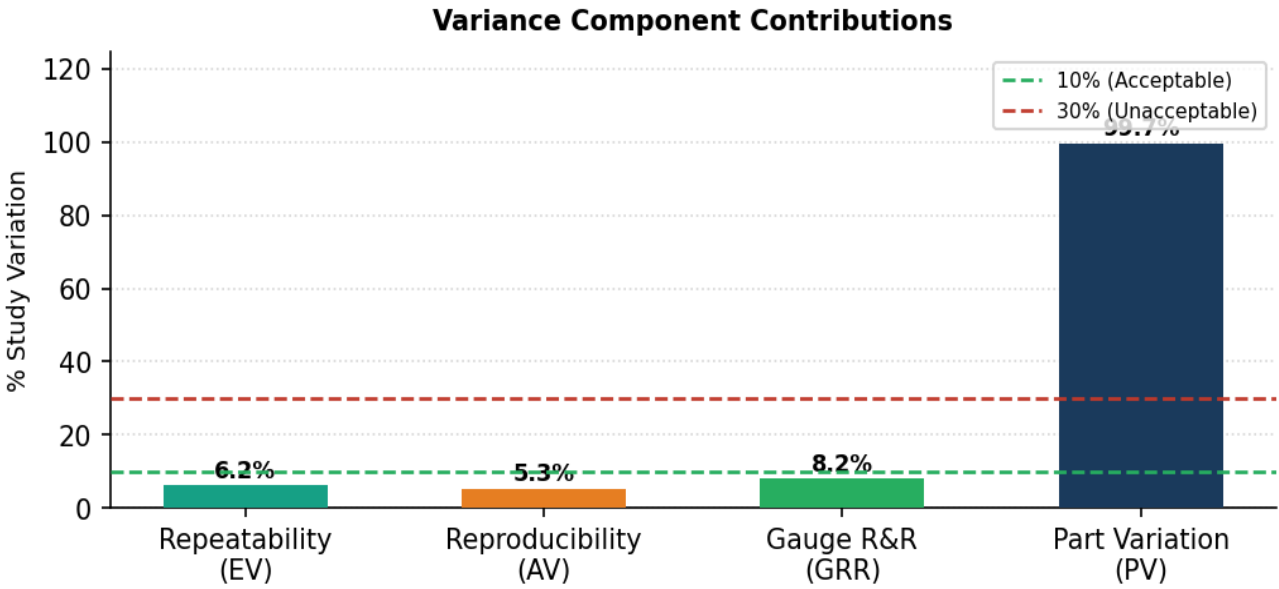
Acceptance Determination

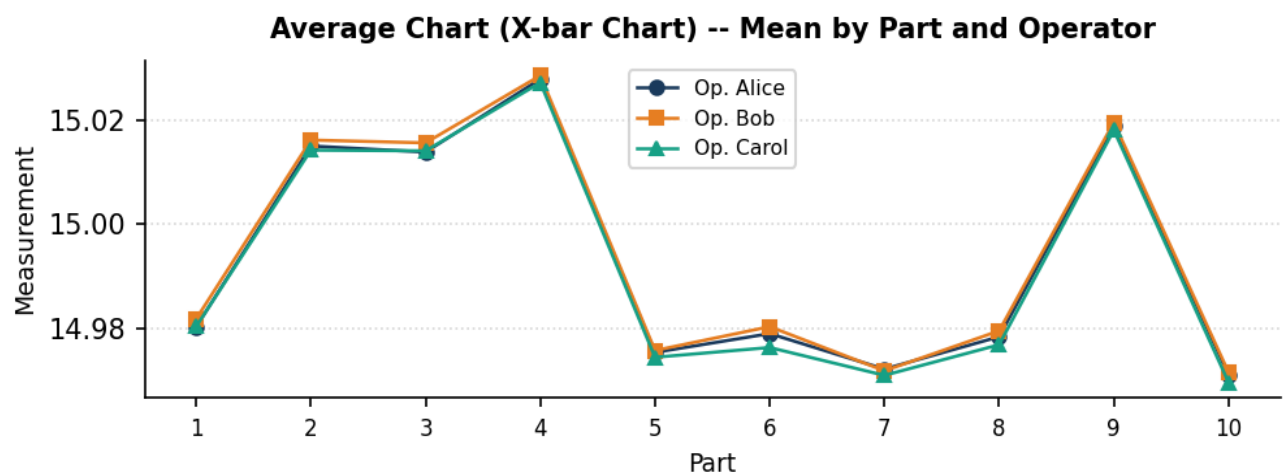
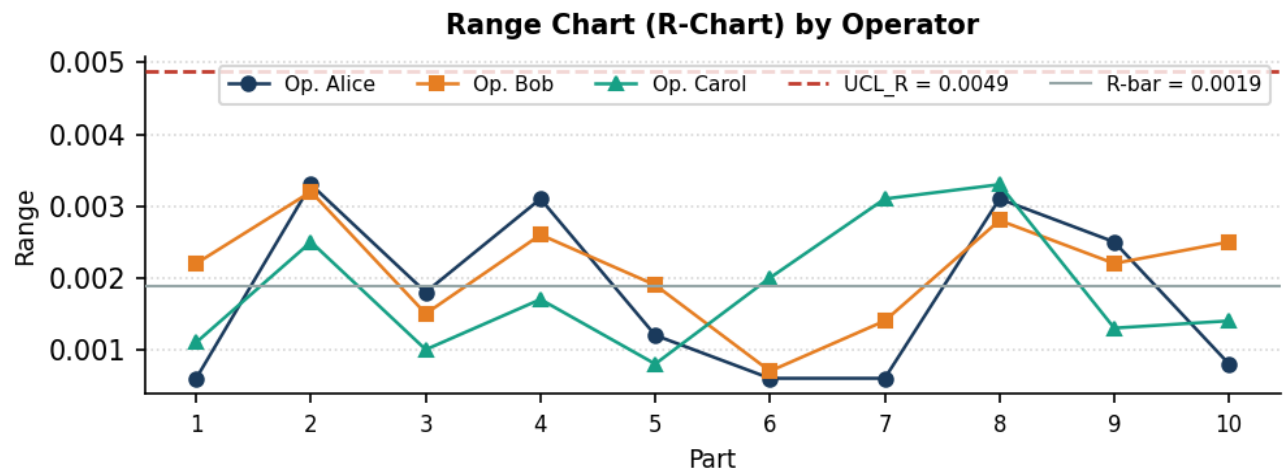
[PASS] ACCEPTABLE (%GRR = 8.2% | NDC = 17)

AIAG MSA Acceptance Criteria Reference

%GRR	Assessment	Recommended Action
<= 10%	ACCEPTABLE	Gauge approved. Measurement system is capable of product acceptance decisions.
10% - 30%	MARGINAL	May be acceptable depending on importance, cost, and customer concurrence. Investigate dominant variation source.
> 30%	UNACCEPTABLE	Do not use for product acceptance. Identify root cause and correct before re-study. Check gage discrimination, comp

Control Charts





The measurement system is **ACCEPTABLE** per AIAG MSA 4th Edition criteria. %GR&R; <= 10%: the gage contributes minimal variation relative to total process variation. This instrument may be used for product acceptance decisions. Continue routine calibration per your control plan.

Generated by `grr_tool.py` -- AIAG MSA 4th Edition Average & Range Method. Study variation expressed as 5.15-sigma (99% coverage). %Study Variation = $100 \times (\text{component} / TV)$. Regulatory basis: 21 CFR Part 820.72 (FDA QSR). This report is a controlled quality record -- retain per document control procedures.